250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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EvolveGraph: Multi-Agent Trajectory Prediction with Dynamic Relational Reasoning

Jiachen Li | Fan Yang | M. Tomizuka | Chiho Choi

Computer Science • Neural Information Processing Systems • 31 March 2020

TLDR This paper proposes a generic **trajectory forecast**ing framework with explicit relational structure recognition **and prediction** via latent interaction graphs among multiple heterogeneous, interactive agents **and** introduces a double-stage training pipeline which not only improves training efficiency **and** accelerates convergence, but also enhances **model** performance. Expand

210 | PDF | View Paper | Save | Cite

Monthly runoff forecasting based on LSTM–ALO model

Xiaohui Yuan | Chen Chen | Xiaohui Lei | Yanbin Yuan | Rana Muhammad Adnan

Environmental Science, Computer Science • Stochastic environmental research and risk... • 22 May 2018

TLDR In the case study, **simulation** results with the historical monthly runoff of the Astor River Basin show that the LSTM–ALO **model** has higher accuracy than that of other **models and** provides an effective method **for** monthly runoff **forecast**ing. Expand

279 | Springer Nature | Save | Cite

Multiscale simulations of complex systems by learning their effective dynamics

Pantelis R. Vlachas | G. Arampatzis | Caroline Uhler | P. Koumoutsakos

Computer Science • Nature Machine Intelligence • 24 June 2020

TLDR It is argued that learning the effective **dynamics** provides a potent novel modality **for** accurately predicting complex systems **and** reduces the computational effort of large-scale **simulation**s by up to

two orders of magnitude while maintaining the **prediction** accuracy of the full system **dynamics**. Expand

161 [PDF] Springer Nature Save Cite

Land Use and Land Cover Change Assessment and Future Predictions in the Matenchose Watershed, Rift Valley Basin, Using CA-Markov Simulation

M. Mathewos Semaria Moga Lencha Misgena Tsegaye

Environmental Science, Geography · Land · 22 September 2022

Land use and **land** cover change (LULC) is known worldwide as a key factor of environmental modification that significantly affects natural resources. The aim of this study was to evaluate the **dynamics**… Expand

37 PDF Publisher Save Cite

Modeling and forecasting the spread of COVID-19 with stochastic and deterministic approaches: Africa and Europe

A. Atangana S. İğret Araz Mathematics, Medicine ·

Advances in Differential Equations · 20 January 2021

TLDR The projection of statistical analysis leads to introducing an extended version of the well-blancmange function to further capture the **spread** with fractal properties, **and** the suggested mathematical **model** is able to predict two to three waves of the **spread** in the near future. Expand

94 PDF Springer Nature Save Cite

Sensitivity of Burned Area and Fire Radiative Power Predictions to Containment Efforts, Fuel Density, and Fuel Moisture Using WRF-Fire

F. Turney P. Saide +9 authors L. Thapa Environmental Science ·

Journal of Geophysical Research: Atmospheres · 11 September 2023

Predicting the evolution of burned area, smoke emissions, and energy release from **wildfires** is crucial to air quality **forecasting** and emergency response planning yet has long posed a significant… Expand

6 PDF Wiley Save Cite

4-D Flight Trajectory Prediction With Constrained LSTM Network

Zhiyuan Shi Min Xu Q. Pan Engineering, Computer Science ·

IEEE transactions on intelligent transportation… · 1 November 2021

TLDR Quantitative analysis demonstrates that the proposed constrained Long Short-Term Memory network for flight **trajectory prediction** outperforms the above-mentioned state-of-the-art **models**, **and** lays a good foundation **for** decision-making in different scenarios. Expand

74 PDF IEEE Save Cite

Short-term load forecasting

G. Gross F. Galiana Engineering, Environmental Science ·

Proceedings of the IEEE · 1 December 1987

This paper discusses the state of the art in short-term load **forecasting** (STLF), that is, the **prediction** of the system load over an interval ranging from one hour to one week. The paper **reviews** the… Expand

1,034 PDF IEEE Save Cite

Vehicle Motion Prediction at Intersections Based on the Turning Intention and Prior Trajectories Model

Ting Zhang Wenjie Song M. Fu Yi Yang Meiling Wang

Computer Science, Engineering · IEEE/CAA Journal of Automatica Sinica · 1 October 2021

TLDR A long short-term memory based (LSTM-based) framework that combines intention **prediction and trajectory prediction** together **and** the credibility evaluation, which is based on the prior trajectories **model**, makes the framework more practical in the real-world applications. Expand

66 IEEE Save Cite

Quantifying 25 years of disease-caused declines in Tasmanian devil populations: host density drives spatial pathogen spread.

C. Cunningham S. Comte +9 authors M. E. Jones

Environmental Science, Biology · Ecology Letters · 27 February 2021

TLDR A pattern-oriented diffusion **simulation** is used to **model** the spatial **spread** of DFTD across the species' range **and** quantify population effects by jointly **modelling** multiple streams of data spanning 35 years, estimating the **wild** devil population peaked at 53 000 in 1996 **and** should level-off within the next decade. Expand

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